11 · Ad exposure is self-selected — the true effect (instrumental variables · CausalPy)

July 13, 2026

> **Runs in the `legacy` environment (pymc<6): `make env-legacy`, kernel `cmp-legacy`.**

The business decision. People who see our ads convert more — but the customers who *end up* seeing our ads are also the ones already leaning toward us (they browse, they search, they get retargeted). So **exposure is self-selected**, and a plain regression of sales on exposure is biased **upward**: it credits the ad for enthusiasm that was already there. We want the *true* causal effect of an extra exposure so we can set the right **budget** — the most we can pay per exposure and still come out ahead.

The problem in one word: endogeneity

When the treatment (ad exposure) is **correlated with the hidden drivers of the outcome** (a customer's unobserved intent), we say exposure is **endogenous**, and no amount of controlling for what we *measured* fixes it — the confounder (intent) is unmeasured. Notebook 05's adjustment tools are powerless here.

The fix: an instrument

An **instrumental variable (IV)** is a third variable Z that nudges the treatment *without* affecting the outcome any other way. Here Z is a randomized **encouragement** — e.g. a serving-priority lottery that makes some customers *more likely* to be shown the ad. Because Z is randomized, the slice of exposure it drives is as-good-as-random, uncontaminated by intent. IV isolates that clean slice and reads the effect off it. A valid instrument needs **four** things:

- **Relevance** — Z actually moves exposure (this one we *can* test: the **first-stage F-statistic**; $F < 10$ is a "weak instrument" danger sign).
- **Exclusion** — Z affects sales *only through* exposure (the lottery itself sells nothing). Untestable from data — defended on design, and stress-tested in Step 6.
- **Exogeneity** — Z is independent of the hidden confounder (guaranteed here by randomization).
- **Monotonicity** — the encouragement never pushes anyone *away* from exposure (no "defiers": nobody is so annoyed by the nudge that they block ads in response). Untestable, but usually plausible for a gentle encouragement — and it is what buys the LATE interpretation below (derived in Step 3).

What IV recovers: a LATE

IV does **not** give the average effect over everyone. It gives the **LATE** (*Local Average Treatment Effect*) — the effect *for the "compliers,"* the customers whose exposure the encouragement actually changed. Always-exposed loyalists and never-exposed sceptics are silent in an IV estimate; report the scope accordingly.

> **On real data.** IV needs a genuine instrument, which is the hard part. Good marketing instruments: randomized **encouragement / intent-to-treat** designs (ship the nudge to a random half), ad-auction or delivery randomness, or a **cost shifter** for the price-elasticity problem in notebook 03. If you can randomize the *encouragement* (not the exposure itself), you have a clean instrument. Notebook **11b** (`11b_endogenous_exposure_real_data.ipynb`) runs this exact playbook on the real Criteo uplift data, where the randomized assignment instruments actual ad exposure — the real-world leg of this encouragement design.

2 · Simulate a ground truth

The **true** exposure→sales effect is **€15**. An unobserved `engagement` drives both exposure and sales (the confounding), so naive OLS overstates it. A randomized `encouragement` (the instrument) nudges exposure without touching sales directly.

The data-generating model — exactly what `dgp.iv_ad_exposure` implements (defaults & seed in `src/cmp/dgp.py`). $n = 3000$; **unobserved** intent U = engagement $\sim \mathcal{N}(0,1)$; randomized encouragement $Z \sim$ Bernoulli$(\frac{1}{2})$; $\sigma(z) = 1/(1+e^{-z})$:

$$\begin{aligned} X = \text{exposure} &\sim \text{Bernoulli}(\sigma(0.3 + 1.1\,Z + 0.8\,U)) && \text{(first stage)} \\ Y = \text{sales} = 50 + 15\,X + 12\,U + \varepsilon, &\qquad \varepsilon \sim \mathcal{N}(0,\,6^2) && \text{(true effect: the 15).} \end{aligned}$$

U appears in **both** equations — $0.8\,U$ pushes exposure and $12\,U$ pushes sales — that is the self-selection, and since U is unobserved no regression adjustment can fix it (OLS is biased upward). All **four** IV conditions are visible by construction: **relevance** ($1.1 \neq 0$ — testable, the first-stage F below), **exclusion** (Z appears nowhere in the sales equation — untestable in real data, stress-tested in step 6), **exogeneity** (Z is a fair coin flip, drawn independently of U — randomization), and **monotonicity** ($1.1\,Z$ only ever *raises* the exposure probability — no defiers).

```
TRUE exposure->sales effect €15  *  first-stage: exposure 56%->77% (shift +21
pp), F = 156
```

	encouragement	ad_exposure	sales
0	1.0	0.0	63.748992
1	1.0	1.0	64.636639
2	1.0	1.0	46.507627
3	0.0	1.0	75.599770
4	0.0	1.0	70.577754

3 · Identify — a valid instrument, and what IV recovers

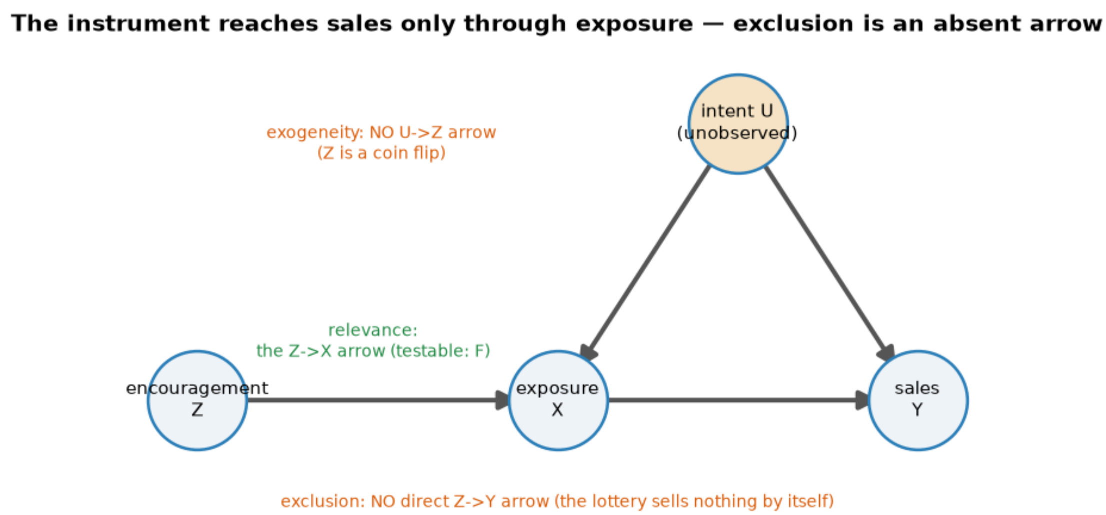
Exposure is **endogenous** — it is correlated with the outcome's hidden driver, Cov(exposure, $12U + \varepsilon) \neq 0$ (the U term specifically, since ε is independent structural noise) — so OLS is biased. The four requirements from the intro, now each named, formalized, and argued in marketing terms:

- **Relevance** — $\pi \neq 0$ in the first stage $X = b_0 + \pi Z + u$: the encouragement genuinely moves exposure. *Testable* — and the only assumption the data can check. The first-stage F is printed in §2; **Step 0** below defines it, derives the F < 10 rule of thumb, and prices what a strong instrument leaks, and §5c degrades the instrument until the estimator comes apart.
- **Exclusion** — the encouragement has **no arrow of its own into sales**. In potential-outcomes notation, writing $Y_i(z, x)$ for customer i's sales if we *set* encouragement to z and exposure to x:

$$Y_i(z, x) \;=\; Y_i(x) \qquad \text{for all } z \in \{0, 1\}:$$

holding exposure fixed, flipping the lottery changes nothing. *Untestable from data* — defended on design (a serving-priority lottery sells nothing by itself; an encouragement *email that shows the product* would violate it), and stress-tested in Step 6. - **Exogeneity / independence** — $Z \perp (U, \varepsilon)$: the instrument is unrelated to the hidden drivers of exposure and sales. Holds here **by randomization** of the lottery. - **Monotonicity** — $X_i(1) \geq X_i(0)$ for every customer i, where $X_i(z)$ is i's exposure when the encouragement is set to z: the nudge never pushes anyone *away* from the ad (no defiers). *Untestable*; plausible for a gentle encouragement, and it is precisely what turns the IV ratio into an interpretable LATE — the derivation below uses it to kill one term.

The DAG states the structural part in one picture — and teaches that two of the assumptions are **missing arrows**:



Read the DAG by its absences. The unobserved intent U points into both exposure and sales — that fork is the endogeneity OLS trips over, and no arrow into U can be measured away. The instrument Z has exactly **one** outgoing arrow, into exposure: exclusion and exogeneity are not things Z *does*, they are arrows Z *does not have* — which is why they cannot be tested from data, only designed for. Relevance is the one arrow Z must have, and the only assumption the data can check.

Who is in the estimate? The four customer types. With $X_i(z) \in \{0,1\}$ the exposure customer i would end up with under encouragement z, two potential exposures give four types — the IV analogue of notebook 01's Persuadable / Sleeping-Dog grid:

	$X_i(1) = 1$ — exposed if encouraged	$X_i(1) = 0$ — unexposed if encouraged
$X_i(0) = 1$ — exposed anyway	**Always-taker** — brand loyalist who seeks the ad out regardless; the lottery changes nothing	**Defier** — so annoyed by the nudge they block the ad they'd otherwise have seen. *Ruled out by monotonicity*
$X_i(0) = 0$ — unexposed otherwise	**Complier** — the lottery tips them into seeing the ad. *The only type IV speaks for*	**Never-taker** — ad-blocker user, barely online; unreachable either way

The LATE theorem (Imbens–Angrist), in two lines. Decompose the **ITT** (intent-to-treat) effect on sales — the average sales difference between encouraged and not-encouraged customers, which *is* the reduced form — over the four types. Randomization of Z makes encouraged vs not comparable; **exclusion** says sales can only react to Z if exposure changed, which silences always- and never-takers ($X_i(1) = X_i(0)$); **monotonicity** deletes the defier term. What survives is the compliers:

$$\underbrace{E[Y \mid Z{=}1] - E[Y \mid Z{=}0]}_{\text{ITT on sales (reduced form)}} \;=\; E\big[(Y_i(1){-}Y_i(0))(X_i(1){-}X_i(0))\big] \;=\; \Pr(\text{complier})\cdot \underbrace{E[Y_i(1) - Y_i(0) \mid \text{complier}]}_{\text{LATE}}.$$

The same decomposition applied to exposure itself gives $E[X \mid Z{=}1] - E[X \mid Z{=}0] = \Pr(\text{complier})$ — **the first stage *is* the complier share** (the +21 pp printed in Step 2: about a fifth of these customers are compliers). Divide the two:

$$\text{LATE} \;=\; \frac{E[Y \mid Z{=}1] - E[Y \mid Z{=}0]}{E[X \mid Z{=}1] - E[X \mid Z{=}0]} \;=\; \frac{\text{reduced form}}{\text{first stage}} \;=\; \hat{\beta}_{\text{Wald}}.$$

So IV does **not** estimate the average effect over everyone: always-takers and never-takers cancel out of the numerator and are literally absent from the estimate — the LATE is the effect *for the customers the encouragement can move.* CausalPy fits the Bayesian version of this ratio: a joint model of (exposure, sales) with correlated errors, *estimating* the endogeneity ρ rather than assuming it away.

> **Modeling note.** Because `ad_exposure` is binary (0/1), this joint MvNormal models the exposure first stage as a **linear-probability (Gaussian)** equation, so the two-equation model is an approximation — the exposure equation's posterior *width* shouldn't be read at face value; the reliable read-outs are its point estimate and the endogeneity ρ. §4b makes this misfit visible on purpose.

Step 0 · The classical read (no likelihood, no priors, no sampler)

Before any Bayesian machinery, ask what a competent analyst would do here **without** it. The answer is not a different analysis: it is the *same estimand* (the complier LATE of §3), identified by the *same four assumptions* (relevance, exclusion, exogeneity, monotonicity), estimated with the simplest tool that is still correct. The causal work lives in the identification argument, not in the machinery — and IV's machinery is old enough (Wright, 1928) that it predates the computer, let alone the sampler.

Three regressions, and the reader must see all three naked, in this order:

$$\underbrace{Y = a_0 + a_1 X}_{\textbf{(1) naive OLS — BIASED}}, \qquad \underbrace{X = b_0 + \pi Z}_{\textbf{(2) first stage}}, \qquad \underbrace{Y = c_0 + \delta Z}_{\textbf{(3) reduced form}}.$$

Regression (1) is the number the business would have used, and §2's data-generating model already told us it is wrong. Regressions (2) and (3) each regress on the **randomized** lottery Z, so each is a clean experiment — but neither is the answer: the first stage is measured in *exposures*, the reduced form in *euros of sales*. Divide one by the other and the units cancel into euros **per exposure**. That is the **Wald ratio**, and it is the whole of IV:

$$\hat{\beta}_{\text{IV}} = \frac{\hat{\delta}}{\hat{\pi}} = \frac{\widehat{\text{Cov}}(Y,Z)}{\widehat{\text{Cov}}(X,Z)} = \frac{\bar{Y}_{Z=1} - \bar{Y}_{Z=0}}{\bar{X}_{Z=1} - \bar{X}_{Z=0}} = \frac{\text{ITT — what the lottery did to sales}}{\text{compliance — what it did to exposure}}.$$

The lottery moved sales by $\hat{\delta}$ and exposure by $\hat{\pi}$; each unit of *lottery-driven* exposure is therefore worth their ratio, and §3's LATE theorem is the argument that this ratio is a causal effect (for compliers). **With a single binary instrument and no controls, two-stage least squares is exactly this ratio** — `cl.iv_2sls` and $\hat{\delta}/\hat{\pi}$ agree to floating point, and the cell below *asserts* that rather than asking you to take it on faith. So why run 2SLS at all? For the two things a bare ratio cannot carry: a **standard error** and the **first-stage F**.

Predict the OLS bias before running the regression. Because we built the DGM, we can compute exactly how wrong regression (1) will be — before it is wrong. With $Y = 50 + 15X + 12U + \varepsilon$, the OLS slope of Y on X converges to the planted effect **plus** whatever part of the error leaks in through the treatment (the **omitted-variable-bias** formula, specialized to this DGM):

$$\hat{a}_1 \xrightarrow{p} 15 + \frac{\text{Cov}(X,\ 12U + \varepsilon)}{\text{Var}(X)} = 15 + 12\,\frac{\text{Cov}(X,U)}{\text{Var}(X)},$$

since ε is independent noise. Even the sign is knowable in advance: the first stage loads on U with $+0.8$, so $\text{Cov}(X,U) > 0$ and OLS must be biased **upward**. The code below re-draws the same simulation *retaining the normally-unobservable* U, evaluates this formula, and only then runs the regressions — theory should call the OLS number before the regression prints it. (That same $\text{Cov}(X,U)$ has a second life in §4, where the Bayesian joint model estimates it as a positive error correlation ρ.)

The covariance choice, and its one-line reason. This is a cross-section of 3,000 customers — no panel, no time series, so no clustering and no HAC. But exposure is **binary**, and by construction the sales variance differs between the exposed and the unexposed arms: the errors are heteroskedastic *by design*. So all three regressions above take **HC1** (heteroskedasticity-robust) standard errors — the right formula for a cross-section that is not homoskedastic. The IV estimate needs more care still. If you hand-roll the second stage — regress Y on the **fitted** $\hat{X}$ from stage one and read `statsmodels`' standard error off that fit — you get a number that is simply *not* the 2SLS standard error, because the residuals are formed against $\hat{X}$ instead of the real X. `cl.iv_2sls` uses the correct residuals. The cell prints both, and (as in notebook 08) the direction of the error is not the one folklore predicts.

Relevance, measured: the first-stage F and the F < 10 rule — invented here, in the classical arm. For a single instrument, the first-stage F is the ordinary F-statistic of regression (2):

$$F \;=\; \frac{R^2/1}{(1-R^2)/(n-2)},$$

where R^2 is the share of exposure variance the instrument explains and $n-2$ counts the residual degrees of freedom: F measures the instrument's **signal in the first stage relative to noise**. **Why is the folklore threshold 10?** With one instrument, a standard approximation says the 2SLS estimator's residual bias, *relative to the OLS bias it is supposed to remove*, is roughly the reciprocal of F:

$$\frac{E[\hat{\beta}_{\text{2SLS}}]-\beta}{E[\hat{\beta}_{\text{OLS}}]-\beta} \;\approx\; \frac{1}{F},$$

so F = 10 means "no more than about a tenth of the OLS bias leaks back in" — a rule with teeth, not folklore. The cell cashes that formula out in euros on *this* instrument. `cl.iv_2sls` returns the F and a `weak` flag (F < 10) precisely so that the check cannot be skipped; **§5c runs the rule backwards** — degrading the instrument on purpose until F falls through 10 — and watches the estimator come apart. (Caveat, from `first_stage_F`'s own docstring: this is the classical **homoskedastic** F. On real, heteroskedastic data prefer the Montiel Olea–Pflueger **effective F**, which corrects the threshold for non-constant first-stage noise, and treat F > 10 as necessary, not sufficient.)

```
OVB theory, computed BEFORE any regression is run:
  OLS ->p 15 + 12*Cov(X,U)/Var(X) = 15 + 8.5 = €23.5

(1) naive OLS    Y~X  [BIASED]: 23.67  [90% CI 22.86, 24.48]  (SE 0.492, HC1
heteroskedasticity-robust, n=3,000)
(2) first stage  X~Z  (pi): 0.2106  [90% CI 0.1828, 0.2383]  (SE 0.0169, HC1
heteroskedasticity-robust, n=3,000)
(3) reduced form Y~Z  (delta): 3.484  [90% CI 2.463, 4.506]  (SE 0.621, HC1
heteroskedasticity-robust, n=3,000)

LATE (2SLS): 16.55  [90% CI 12.74, 20.35]  (SE 2.31, 2SLS (homoskedastic),
n=3,000)
THE WALD IDENTITY   delta/pi = 3.4844 / 0.2106 = €16.5470   vs   2SLS = €16.5470
  -> identical to floating point. 2SLS *is* the Wald ratio here; the machinery
buys the SE and the F.
```

```
STANDARD ERROR - three formulas for the SAME €16.5 point estimate:
  hand-rolled 2nd-stage OLS (residuals vs the FITTED Xhat - WRONG) :  2.95
  proper 2SLS              (residuals vs the REAL X)              :  2.31  <-
cl.iv_2sls
  2SLS + HC1 sandwich      (heteroskedasticity-robust)            :  2.31
  -> the hand-rolled SE is x1.27 the correct one - and note the DIRECTION: too
     WIDE, not too narrow. Its residual y - b*Xhat carries an extra beta^2*Var(X
- Xhat) term
     (a non-negative quantity), so it charges the estimate for first-stage
scatter that is not
     sampling error at all. Robustness, by contrast, barely moves it (x1.00):
the
     homoskedastic 2SLS interval printed above is safe to report as-is.

RELEVANCE - first-stage F = 156   (cl.iv_2sls weak flag: strong, F >> 10)
  1/F cashed out: OLS is off by €+8.7; the rule says ~1/F of that bias
  leaks back into 2SLS -> €8.7 / 156 ~ €0.06 per exposure. Negligible - which is
why §5b's
  residual bias (an order of magnitude larger) cannot be blamed on the
instrument.

GRADE vs the planted truth (€15/exposure):
  (1) naive OLS  € 23.7  off by €+8.7   truth is OUTSIDE its 90% CI [€22.9,
€24.5]
      - the tightest interval in the notebook (SE €0.49) around a number that is
€8.7 wrong.
        Precision is not accuracy: a confident interval on a biased estimator is
a confident lie.
  (4) 2SLS       € 16.5  off by €+1.5   truth is INSIDE its 90% CI [€12.7,
€20.4]
      - the self-selection wedge the instrument removed: €7.1 per exposure.

What this 90% confidence interval does NOT say: that there is a 90% probability
the true effect lies inside it. It is a property of the *procedure* - 90% of
intervals built this way would cover the truth across repeated samples. This
interval either contains the truth or it does not. To get a probability *about
the effect itself* - the thing a go/no-go rule like P(lift > cost) >= 0.9
actually needs - you need a posterior. That is what the Bayesian section adds.
```

Read-out — the classical answer, in business terms. The naive regression prices an exposure at the number printed above; 2SLS prices it several euros lower, and the difference between them *is* the self-selection — euros of "effect" that were never caused by the ad, only correlated with the enthusiasm of the people who ended up seeing it. Budget on the first number and you overpay on every exposure bought; §6 turns the second into a bid cap. Three details are worth pausing on, because each is a lesson the Bayesian section will inherit rather than replace:

1. **The theory called the OLS number before the regression ran.** The omitted-variable-bias prediction and the fitted OLS slope (both printed above) agree to a fraction of a euro.

The bias is not bad luck or a small sample — it is a *computable* consequence of the DGM, and on real data the same formula tells you the *sign* of your bias even when you cannot compute its size.

2. **The naive OLS interval is the tightest in this notebook and it misses the truth entirely.** It is off by more than half the effect, with a standard error small enough to look authoritative. This is the single most useful thing in Step 0: **precision is not accuracy**, and no amount of extra data would have saved it — more customers would only shrink that interval around the same wrong number.
3. **The first-stage F is a precondition, not a footnote.** F in the hundreds is what entitles us to read the ratio at all; the 1/F cash-out prices the residual leakage at a few cents. §5c shows what the same estimator does when that number falls below 10 — it is not a graceful degradation.

Notice what has *not* happened yet. No likelihood was written down, no prior was chosen, no sampler ran. The identification — a randomized instrument, excluded and monotone — did all the causal work, and two least-squares slopes and a division did all the estimation. **This is the honest baseline against which the Bayesian section must justify itself**, and §5x will hold it to that.

The guardrail — what that confidence interval does *not* say (printed above by `cannot_say()`, in the same words in every notebook of this cookbook): it is **not** a 90% probability that the true exposure→sales effect lies inside it. A confidence interval is a property of the *procedure* — 90% of intervals built this way would cover the truth across repeated samples — and this particular interval either covers the truth or it does not. Here is exactly where that bites: §6 must answer *"what is the probability an exposure is worth more than the €15 we would have to bid for it?"* That sentence asks for a probability **about the effect itself**. The classical apparatus cannot supply one — not because its arithmetic is worse (it is not: 2SLS lands nearer the truth than the posterior will), but because that quantity does not exist in its vocabulary. It exists in a posterior. That, and not a better point estimate, is what the next section is for.

4 · Estimate — the Bayesian IV model

Step 0 already produced a causal number: the Wald ratio, which is 2SLS, with an honest confidence interval and a first-stage F that certifies the instrument. What follows does **not** change the identification by a comma — same instrument, same four assumptions, same complier LATE. It changes the apparatus for expressing uncertainty about that number, and it estimates one object the classical route structurally cannot hand you: the **endogeneity itself**.

The Bayesian IV fits the first stage and the outcome equation **jointly**, allowing their errors to be correlated:

$$\begin{pmatrix} X_i \\ Y_i \end{pmatrix} \sim \mathcal{N}_2\left(\begin{pmatrix} b_0 + \pi Z_i \\ \beta_0 + \beta X_i \end{pmatrix},\ \Sigma \right), \qquad \Sigma = \begin{pmatrix} \sigma_X^2 & \rho\,\sigma_X \sigma_Y \\ \rho\,\sigma_X \sigma_Y & \sigma_Y^2 \end{pmatrix},$$

where the off-diagonal ρ **is** the endogeneity: the correlation between the two equations' errors that the hidden U induces and that OLS ignores. Priors: weakly-informative $\mathcal{N}(0, 50^2)$ on the regression coefficients (the code comment explains why the library default is too tight here) and an **LKJ prior** on Σ — a standard weakly-informative prior over correlation matrices that lets the data decide how strongly the two equations' errors move together (i.e. how big ρ is), rather than fixing it in advance. The β in the second row is the same β Step 0 estimated by 2SLS: one estimand, two machines.

The one genuinely new object: ρ. 2SLS *removes* the endogeneity by projection — it never names it, and it has no ρ to report. The joint model *estimates* it, and the posterior for ρ turns out to be the Bayesian image of exactly the $\mathrm{Cov}(X, U)$ that Step 0's omitted-variable-bias formula priced in euros. A positive ρ **is** the reason the OLS number sat above the IV number; here we get an interval on it. That, plus the ability to ask $P(\beta > \text{cost})$ in §6, is what the sampler is being paid for — and §5x audits whether it earned its fee.

```
Step 0, recalled:   OLS (biased) €23.7   *   Wald = 2SLS €16.5 (90% CI [€12.7,
€20.4])
Bayesian IV         €17.7  (true €15)  90% credible interval [€14.3, €21.1]
IV convergence: max r-hat 1.010 - min ESS 903 - divergences 0


PRIORS, PRICED - the library default vs the weakly-informative override:
  CausalPy 0.8.1 default    € 16.6  90% [€15.3, €17.9]  width €2.6  -> true €15
is OUTSIDE - it MISSES the truth
  N(0, 50^2) [used above]   € 17.7  90% [€14.3, €21.1]  width €6.8  -> true €15
is INSIDE
  the default band is 2.6x TOO NARROW (the classical Wald 90% CI, an independent
yardstick, is €7.6 wide).
  A prior centred on the 2SLS point with sigma=1 does not let the data speak: it
hard-codes
  the answer and then reports false confidence in it. That is why sigmas=[50,
50] is passed.

Estimated endogeneity  rho(exposure err, sales err) = +0.22 [90% +0.10, +0.33]
  - nonzero rho is why OLS (€23.7) sits above IV (€17.7); it is the Bayesian
  image of the same Cov(X,U) Step 0's OVB formula priced at +€8.5. No classical
  estimator in Step 0 has a rho to report: 2SLS projects the bias away without
naming it.
```

Reading the sampler's two health checks. The Bayesian IV is fit by simulation — **MCMC**, the algorithm that draws samples from the posterior — so PyMC prints two numbers that say whether the sampler actually converged, and it will warn loudly if they look loose. **R-hat** compares the variance *within* each MCMC chain to the variance *across* chains; it should sit at ≈1.00, and **≤ 1.01 is the usual pass bar**. **ESS** (effective sample size) is roughly how many *independent* draws the autocorrelated chain is worth — a few hundred is ample for a posterior mean or interval. **Divergences** — steps where the sampler broke down — should be **0**. Under this notebook's FAST teaching profile the chains are deliberately short (only 200 draws × 2 chains to keep it quick), so R-hat can drift upward and ESS can fall toward ~100, tripping PyMC's *"problems during sampling"* and *"effective sample size … smaller than 100"* notices. Those would be **artifacts of the small draw count, not a broken estimate** — and they flag *how well we sampled the posterior*, never whether the IV logic is right. Here, at FULL (4 chains, 1,000 draws), the `IV convergence:` line above prints a clean pass — read the R-hat, ESS and divergence numbers off it rather than off this paragraph. (The `90% credible interval` printed on the same line is the Bayesian analogue of a confidence interval: given the model and priors, there is a 90% posterior probability the true effect lies inside it — which is what lets us later say "the whole posterior sits above €10." The real backstop that the number is trustworthy is the multi-seed recovery-and-coverage check in §5b, not any single run's sampler line.)

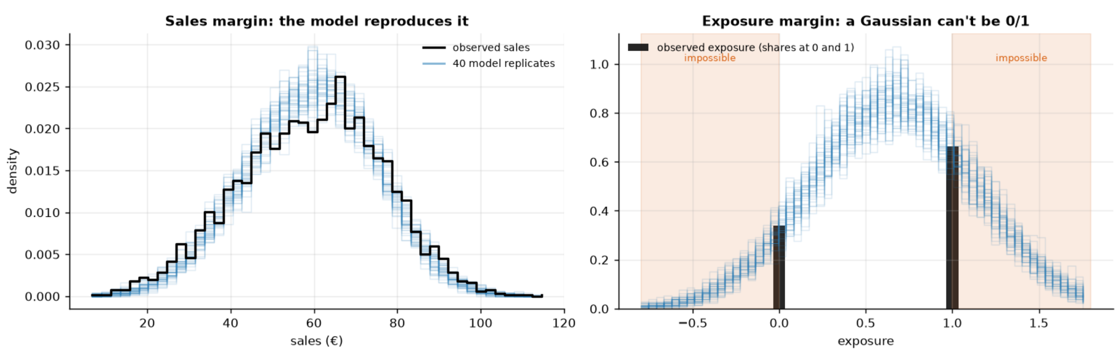
4b · Posterior predictive check — what the joint Gaussian gets right, and what it can't

Before leaning on the estimate we ask the fitted model to **reproduce the data it just saw**. For each of ~40 posterior draws we simulate a full replicate dataset from the fitted joint model,

$$(Y_i^{rep},\, X_i^{rep}) \;\sim\; \mathcal{N}_2\!\left(\begin{pmatrix} \hat{\beta}_0 + \hat{\beta}\, X_i \\ \hat{b}_0 + \hat{\pi}\, Z_i \end{pmatrix},\ \hat{\Sigma} \right),$$

(one $(\hat{\beta}_0, \hat{\beta}, \hat{b}_0, \hat{\pi}, \hat{\Sigma})$ per draw) and overlay the replicated margins on the observed ones. The **sales margin should pass** — that is what licenses reading the causal β off the outcome equation. The **exposure margin cannot pass**: observed exposure is 0/1, and a Gaussian equation will cheerfully produce "exposures" of -0.3 or 1.4. We run it anyway, because that visible failure *is* the Step-3 modeling note, shown rather than footnoted.

```
Sales margin: replicate sd €15.7 vs observed €17.1, means €59.3 vs €59.3
  - close, with a mild under-dispersion (the joint approximation shows up
  even here, faintly): good enough to read beta off the outcome equation.
Exposure margin: 32% of replicated 'exposures' land outside [0,1]
  - impossible values, the linear-probability first stage made visible.
  Read that equation's point estimate and rho, never its width.
```



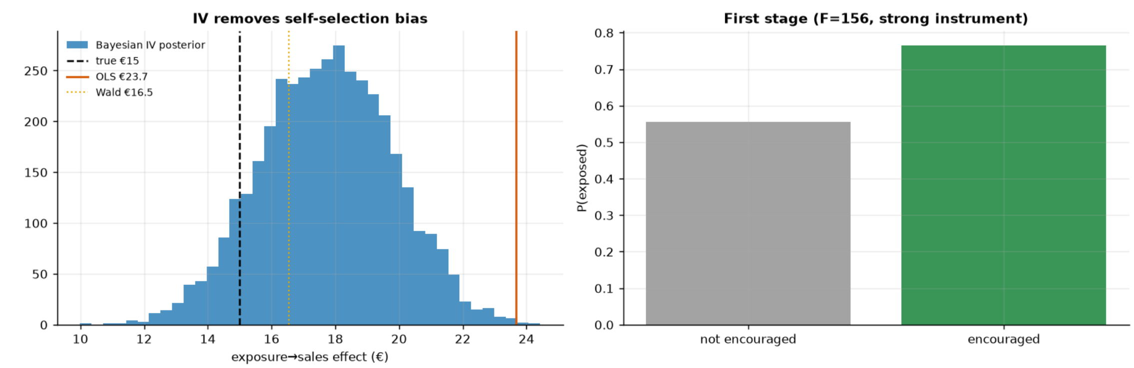
Read-out. *Left*: the replicate sales curves trace the observed distribution well — centred right, roughly the right width, if a hair narrower in the shoulders (the printed sd comparison quantifies it — even the "passing" margin carries a faint fingerprint of the approximation). That is a good enough fit to entitle us to read the causal β (and its interval) off the outcome equation. *Right*: the replicated "exposures" are smooth bell curves, and a substantial share of their mass sits in the shaded regions below 0 and above 1 — exposure values no customer can have (the printed line quantifies it). This is not a surprise to fix; it is the **linear-probability approximation made visible**, and it is exactly why §5b will find a small residual bias and mild under-coverage, and why the Step-3 modeling note told us to use the exposure equation's point estimate and ρ but never its posterior width. One diagnostic now carries what were two apologetic footnotes.

5 · Validate — IV removes the self-selection bias

Naive OLS should sit well *above* the truth; the Wald ratio and the Bayesian IV should both **remove that bias and land close to €15**, with the IV posterior's 90% interval **covering the truth**. The gap between OLS and IV **is** the self-selection bias the instrument removes. (We give the Bayesian IV weakly-informative priors so its interval reflects genuine sampling uncertainty — roughly the frequentist Wald width — rather than the artificially tight band CausalPy's default 2SLS-centred priors would produce.)

One honest note on LATE (and who the compliers are). IV recovers the effect for *compliers* — the customers whose exposure the encouragement actually moves — not the whole population. Under monotonicity that group is exactly the **first-stage shift: +21 pp (56% → 77%)**, the estimated complier share. In *this* simulation every customer's true effect is €15 (the effect is homogeneous), so **LATE = ATE** and grading the IV against €15 is a fair test. On real data, where the effect varies across customers, IV would recover the complier-weighted LATE, which generally ≠ **the ATE** — and that is the right quantity for a budget call, since it answers *"what does an extra exposure do for the customers we can actually move?"*

```
self-selection bias removed: OLS €23.7 -> IV €17.7 (true €15)
```

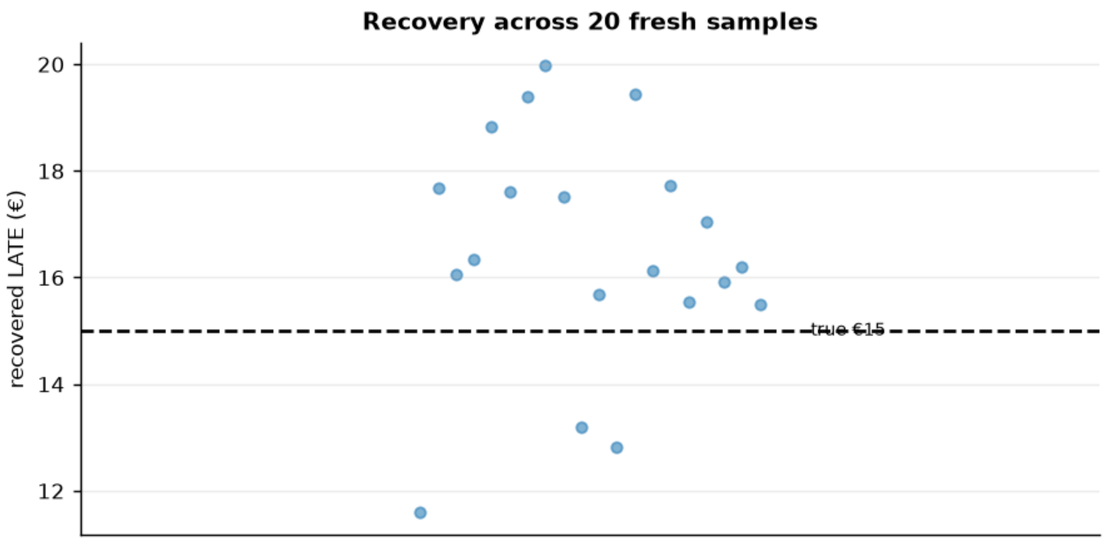


How to read this. *Left* — three vertical lines tell the whole story: **OLS (orange) sits well above the truth** (it credits the ad for pre-existing intent), while the **Bayesian IV posterior (blue) and the Wald ratio (gold) both drop back down toward the true €15**. The IV point carries a small upward finite-sample bias (a couple of euros above the truth on this seed; §5b measures the average across seeds), but its posterior — now honestly wide, thanks to weakly-informative priors — **covers €15** (just inside the lower edge); the artificially tight default-prior band would have excluded it. The horizontal distance from the orange line to the blue posterior *is* the self-selection bias — the euros of "effect" that were never causal. If you budgeted on OLS you'd over-spend on every exposure by exactly that gap. *Right* — the first stage: the encouragement clearly moves exposure (a big jump in P(exposed)), which is why the instrument is **strong** (F $\gg$ 10). A weak first stage here would make the whole IV estimate unstable, so this panel is a precondition, not a footnote.

5b · Recovery across many seeds — unbiased *and* calibrated?

A single sample can land a little off; the real test is whether the estimator is **centred on the truth over repeated samples** and whether its 90% interval **covers** the truth at the stated rate. We refit on many fresh samples (a small fast fit each; their sampler chatter is silenced below, and we read only the aggregate bias/coverage tallies) and check both.

```
IV across 20 seeds: mean €16.5 (true €15)  bias +1.5  sd €2.1
  *  90% interval covers truth in 16/20 seeds.
The committed seed (37) lands high (€18); across seeds the mean is €16.5
  - a small finite-sample bias of +1.5 € (~10%) toward OLS (the next cell asks
  whether it lives in the identification or in the model). The 90% interval
  covers €15 in 16/20 seeds (80% vs the 90% target - modest under-coverage,
  consistent with that bias); the budget verdict below never hinges on the
  last euro, which is why the ranking and the cap-style decision survive it.
```



Is the residual bias the identification's fault, or the model's? The Wald ratio *is* the IV identification with nothing else attached — two least-squares slopes and a division; no priors, no likelihood, no sampler. Re-running it on fresh draws at growing n therefore isolates what finite samples do to the identification itself. Whatever bias it does **not** show must belong to the Bayesian machinery around it — and §4b already fingered the suspect: the Gaussian first stage.

```
closed-form Wald across 30 seeds (identification only, no model):
  n =  1,000:  mean € 14.4  (bias -0.6)   spread (sd) €3.71
  n =  3,000:  mean € 15.1  (bias +0.1)   spread (sd) €2.53
  n = 10,000:  mean € 15.0  (bias +0.0)   spread (sd) €1.43
The Wald column stays pinned near €15 at every n - its deviation is within
sampling noise while the spread shrinks like 1/sqrt(n). The identification is
clean; the posterior-mean bias above therefore belongs to the Bayesian model's
finite-sample behaviour (the Gaussian first stage §4b exposed, plus
posterior-mean-of-a-skewed-posterior at modest draw counts), not to the IV
```

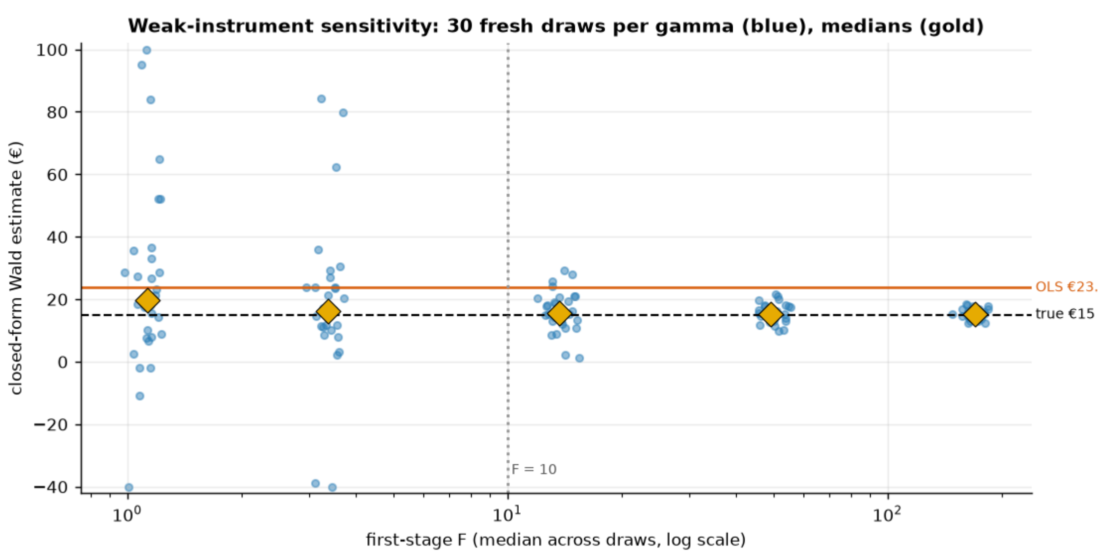
```
logic. Step 0 already priced the pure-IV leakage, via the 1/F rule, at a few
cents.
```

5c · What if the instrument were weak? (the headline pitfall, demonstrated)

Everything so far used a **strong** instrument: the encouragement moves exposure by roughly a fifth, and the first-stage F printed in Step 0 sits in the hundreds. Step 0 defined that F, derived the F < 10 rule from the $1/F$ bias approximation, and cashed the rule out on *this* instrument at a few cents of leaked OLS bias per exposure. Now we run the rule **backwards**. The caveat "weak instruments are dangerous" deserves better than assertion, so we degrade the instrument on purpose, watch F fall through 10, and watch the estimator come apart.

The experiment. `dgp.iv_ad_exposure` (src/cmp/dgp.py) hard-codes the encouragement coefficient at 1.1, so the cell below re-parameterizes the same six-line DGP with the first-stage strength γ as a dial — $X \sim \text{Bern}(\sigma(0.3 + \gamma Z + 0.8U))$, $Y = 50 + 15X + 12U + \varepsilon$ — and sweeps γ from this notebook's strength down to nearly-useless. For each γ we draw many fresh datasets and compute the **closed-form Wald only** (two `lstsq` calls per draw — the failure we are demonstrating belongs to the *identification*, not to any Bayesian model, so no MCMC is needed and the cell stays seconds-fast). This is Step 0's estimator, and Step 0's estimator alone, under stress.

```
  gamma  1.1: median F  169.6  median Wald € 15.2  90% range [12.7, 18.0]
  gamma  0.6: median F   49.3  median Wald € 15.3  90% range [10.7, 20.6]
  gamma  0.3: median F   13.7  median Wald € 15.6  90% range [5.0, 27.0]
  gamma 0.15: median F    3.4  median Wald € 16.3  90% range [-20.4, 71.9]
  gamma 0.08: median F    1.1  median Wald € 19.8  90% range [-6.9, 90.2]
At F ~ 1.1 the 90% range of the IV estimate spans €-7 to €90 - far wider than
the €8.7 OLS-vs-truth gap it was meant to fix, and the median drifts toward OLS
(€23.7): a weak instrument is worse than none - shown, not asserted.
(3 extreme point(s) clipped to the plot window.)
```



How to read this. Each blue column is the *same estimator on the same market* — only the instrument's strength changes. At the right edge (F in the hundreds, this notebook's regime) the Wald estimates hug the true €15. Moving left, the spread widens smoothly until, below the dotted F = 10 line, it explodes: single draws can land below zero or at multiples of the truth, and the gold *median* drifts up toward the orange OLS line. The mechanism is the 1/F formula read right-to-left — with a nearly-irrelevant instrument the Wald denominator is sampling noise around zero, so the ratio inherits OLS's bias *and* enormous variance. That is the precise sense in which **a weak instrument is worse than none**: OLS is wrong by a known-ish €9 with a tight interval, while a weak IV can be wrong by tens of euros in either direction *while wearing the costume of a causal estimate*. Lecture takeaway: check F before believing any IV number, and if F is small, walk away rather than report the ratio.

5x · Point estimate vs posterior — what the Bayesian layer actually bought

Step 0's 2SLS and Step 4's posterior target the **same estimand** (the complier LATE) under the **same four assumptions**, on the **same 3,000 customers**. Only the apparatus differs. The table below lays them side by side against the planted truth, and then asks each one the question §6 actually needs answered.

```
SAME estimand (the complier LATE), SAME four assumptions, SAME 3,000 customers.
Planted truth = €15/exposure.  (Rows 1-2 carry 90% CONFIDENCE intervals,
 row 3 a 90% CREDIBLE interval - printed in the same column, but NOT the same
object.)

estimator (€ per exposure)           est     5%    95%  +/-half  |err|
naive OLS * HC1        [BIASED]     23.7   22.9   24.5    0.8    8.7
Wald ratio = 2SLS * Step 0          16.5   12.7   20.4    3.8    1.5
Bayesian IV posterior * Step 4      17.7   14.3   21.1    3.4    2.7

LOCATION - the two causal estimators agree with each other to within €1.2, and
both
  sit ABOVE the truth (2SLS +1.5, Bayes +2.7). Both intervals COVER €15.
  Closer to the truth on this seed: 2SLS (classical)  (|err| €1.5 vs €2.7).
  Bayes does NOT win the point estimate here. The naive OLS row is the outlier -
€+8.7 off,
  and its interval, the tightest of the three (+/-€0.8), misses the truth by
miles.
WIDTH  - +/-€3.8 (2SLS) vs +/-€3.4 (posterior): the same scale, and that is by
design -
  §4's weakly-informative priors were chosen precisely so the data, not the
prior, sets the
  width. The posterior bought no precision, and was not supposed to.

P(effect > c) - a probability ABOUT the effect, at three candidate bid caps:
  c (€ per exposure)       classical   Bayesian IV
  10                     not defined         1.00
  15                     not defined         0.90
  18                     not defined         0.45
  ('not defined' is not a dodge. A confidence interval carries no probability
```

```
about a
   hypothesis, and neither would a bootstrap of the Wald ratio: the share of
resamples
   above c is a frequency over hypothetical datasets, not a belief about beta.
§6's rule
   -- bid up to c while P(effect > c) >= 0.90 -- needs the right-hand column and
nothing else.)

And the endogeneity itself: rho = +0.22 [90% +0.10, +0.33] - an interval on the
very bias
  that 2SLS silently projects away. Step 0 has no rho to report, at any width.
```

The honest verdict — what the posterior bought, and what it did not.

1 · They agree, and they agree in the same direction: both a little high. As the table prints, 2SLS and the Bayesian posterior mean land within a euro or two of each other, and *both* sit above the planted €15. The classical estimator is not a strawman here — it is a competent estimator of the same quantity, and on this seed it is the **closer of the two to the truth**. Say that plainly: **Bayes did not win the point estimate.** Nor is either miss mysterious; the notebook has already diagnosed both halves. §5b re-ran the closed-form Wald on fresh draws at growing n and found it **pinned on €15** — so the identification is clean and the classical arm's residual is ordinary sampling noise on one draw. The posterior mean's extra drift is the Bayesian model's own: §4b showed its Gaussian first stage cheerfully producing "exposures" of -0.3 and 1.4 for a 0/1 variable, and §5b measured the small upward bias and mild under-coverage that approximation induces. **The Bayesian machinery introduced a modelling error the two `lstsq` calls of Step 0 could not make.** That is the ledger, and it does not flatter the sampler.

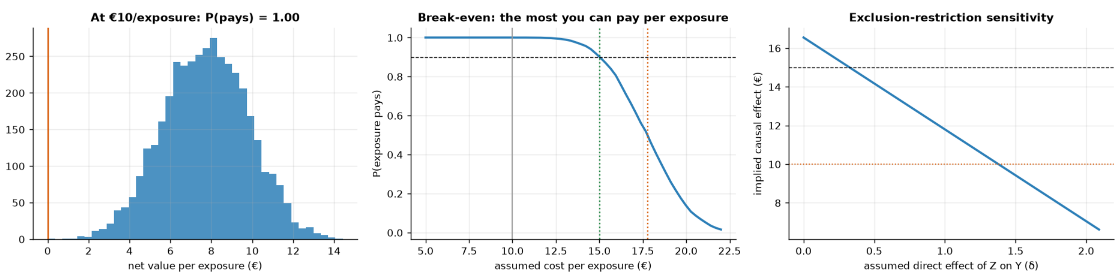
2 · What the posterior *does* buy — the two blocks at the bottom of the print-out. Look at the `P(effect > c)` column. The classical row reads *not defined*, and that is the literal truth rather than a rhetorical concession: the frequentist apparatus attaches no probability to a hypothesis about a fixed parameter, and — this is the part people get wrong — **bootstrapping the Wald ratio would not fix it either**. The share of bootstrap resamples landing above €15 is a frequency over hypothetical *datasets*, not a probability about the *effect*. Yet §6's bid-cap rule is nothing but such a probability: *bid up to c while* $P(\mathit{effect} > c) \geq 0.90$. The entire euro decision — the €15 cap, the €10 plan's headroom, the exclusion stress test — is computed from the right-hand column and could not have been computed from the left. Second, ρ: 2SLS removes the endogeneity by projection and never names it, while the joint model returns a posterior interval on it. On real data, where there is no answer key, a ρ that is credibly positive is the evidence that the naive number *was* inflated — evidence the classical arm cannot produce.

3 · What Bayes did NOT buy here, specifically. Not a better point estimate (2SLS is nearer the truth). Not a tighter interval (the widths match by design — and the *tightest* interval in this notebook belongs to the biased OLS, which is the whole point of §5x). Not protection from model error (it *created* one, in §4b's linear-probability first stage, which Step 0's estimator sidesteps by never modelling X at all). Not the weak-instrument diagnostic — the F is a classical object, and the posterior would have gone on producing a beautifully smooth, entirely fictitious credible interval as F fell through 10 (§5c). What it bought is **the decision**: a probability statement about the effect, propagated end-to-end into a bid cap in §6, plus a coherent read-out of the endogeneity that caused the problem in the first place. On a problem this well-instrumented and this data-rich, that is the whole of the win — and it is enough, because it is precisely the thing the CMO asked for.

6 · Decide, in euros — and the exclusion-restriction stress test

Budget on the *causal* per-exposure value (IV), not the inflated OLS. But the exclusion restriction is untestable, so we ask: **what if the encouragement had a small direct effect on sales** (say the lottery email itself advertises)? We subtract a hypothetical direct effect δ from the reduced form and watch the implied causal estimate move — the honest bound on how much a leak would distort the number.

```
At €10/exposure: causal net €7.7  *  P(pays) 1.00
  -> BUY exposure (to the complier margin)
  P(pays)=1.00 is saturated, not assumed: the whole posterior
  [€14.3, €21.1] sits above €10.
Break-even sweep: exposure clears the 90%-probability bar up to €15.0/exposure
  (the budget cap); at €17.8 it is a coin flip.
  So the €10 plan has ~€5.0 of headroom per exposure before the call gets tight.
Exclusion stress: the BUY call survives unless the instrument's direct effect
  exceeds delta~1.4 (out of a €3.5 reduced form) -
  trusting OLS (€24) would overstate the value and overspend.
  (the corrected line is a point-estimate bound - it re-solves the Wald ratio
  at posterior means; carry the IV interval's width alongside it when briefing
  the cap.)
```



Read-out. Three numbers make the budget case, one per panel. *Left*: at the current €10/exposure the **causal** net value is ≈ €8/exposure with the entire posterior above zero — a saturated BUY, valid for the complier margin the instrument identifies. *Middle*: the sweep turns the posterior into a procurement cap — the 90%-**probability** bar (a posterior quantity: the classical arm of §5x has no such object to offer) holds up to ≈ **€15/exposure**, the coin-flip point is ≈ €17.8, so the €10 plan has ~€5 of genuine headroom. *Right*: the stress test bounds the untestable worry — the BUY survives unless the lottery email itself drives more than δ ≈ €1.4 of sales directly, which against a €3.5 total reduced form would be a large leak (a plain encouragement email with no product content makes that implausible; an email that doubles as an ad does not — design the instrument accordingly). The management contrast to land: budgeting on OLS's €24 would sanction paying up to €24 for something causally worth ≈ €15 — a €9 gap against the simulator's answer key. On real data the truth is unknown, so the wedge you'd actually avoid is OLS − IV ≈ €6/exposure (the IV prices the exposure at ≈ €17.7); either way, times every exposure bought, that is the price of mistaking correlation for incrementality — the cell below prices that same error at campaign scale (≈ €5.9M).

The one-paragraph decision

> **What I'd tell the CMO.** Buy the exposure at the current €10: the causal value per exposure (the IV posterior above) sits comfortably clear of cost — P(pays) is saturated — and the break-even sweep caps procurement at ≈ €15 per exposure at 90% posterior probability, coin-flip near €18, so the €10 plan carries roughly €5 of headroom. **Scope:** that number is earned on the *complier margin* — the ≈ 21 pp of customers the encouragement actually tips into seeing the ad — and says nothing about always-exposed loyalists; that is the right scope for this call, because a budget buys *incremental* exposure. **Fragility, priced:** the BUY survives a direct-effect leak of the lottery into sales up to $\delta \approx$ €1.4 of the €3.5 reduced form (a large leak for a content-free lottery email), and survives the small finite-sample bias §5b measured; it would *not* survive a weak instrument — which is why F ≈ 156 was checked before anything else (§5c shows the crash we avoided). **The mistake we didn't make:** budgeting on the naive €23.7 would sanction paying several euros of non-causal value on every exposure bought — the cell below prices that error at campaign scale, and the JSON carries every number a deck needs.

```
Campaign-scale cost of the naive number: an OLS-calibrated cap sanctions
  ~ €5.9 of non-causal value per exposure; at 1,000,000 exposures/quarter that
  authorizes ~ €5.9M per quarter for value that does not exist
  (€2.5M-€9.3M against the IV 90% interval).
```


```
{
  "true_effect": 15.0,
  "ols": 23.67,
  "wald": 16.55,
  "iv_mean": 17.73,
  "iv_ci90": [
    14.34,
    21.15
  ],
  "first_stage_F": 156.2,
  "complier_share_pp": 21.1,
  "rho_mean": 0.218,
  "cost_per_exposure": 10.0,
  "budget_cap_c90": 15.02,
  "coinflip_c50": 17.77,
  "exclusion_break_delta": 1.43,
  "multiseed_bias": 1.5,
  "multiseed_coverage": "16/20",
  "naive_overpay_eur_per_quarter_at_1M": 5940601
}
```


7 · Caveats

- **LATE, not ATE.** IV speaks only for **compliers** — customers the encouragement moves (the Step-3 table shows who is silent). Always-exposed loyalists and never-exposed skeptics may respond differently; report the scope.
- **Weak instruments are dangerous.** A small first stage (low F) inflates both variance

and bias — §5c demonstrated the crash: below F $\approx$ 10 the estimate's spread explodes and its median drifts back toward OLS. Check F first (on real data, the Montiel Olea–Pflueger *effective* F), and if it is small, walk away rather than report the ratio.

- **Exclusion is untestable.** We formalized it ($Y_i(z,x) = Y_i(x)$), stress-tested it in step 6, and priced the leak that would flip the call; defend it on design, not data.
- **Monotonicity (no defiers).** IV's LATE interpretation assumes no one is pushed *away* from exposure by encouragement — usually plausible, worth stating (it deleted the defier row in Step 3's derivation).
- **The joint Gaussian is an approximation.** §4b showed the exposure margin failing by construction (a 0/1 variable modeled linearly); §5b measured the resulting small bias and mild under-coverage. Read the exposure equation's point estimate and ρ, never its posterior width.
- **Real data next.** Notebook **11b** replays this playbook on the Criteo uplift dataset — randomized assignment as the instrument for actual ad exposure — where F, the complier share, and the budget cap must be earned from the data rather than planted in it.